

High Performance Clusters

1

What is a High Performance Cluster (HPC)?

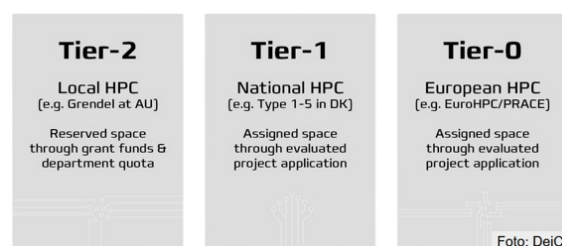
- Cloud providers offer near infinite compute resources, but at a high cost.
- HPCs are a cost-effective alternative, offering access to powerful computing resources.
- **HPCs exist worldwide**, and many individuals have access through institutions or public resources.
 - University students often have access to local HPCs.
 - Public HPC resources are available through application, like the EuroHPC initiative in the EU.



2

Understanding HPC Tiers

- HPCs are categorized into tiers based on their capabilities:
 - **Tier-0:** European centers with petaflop or hexascale machines (largest applications).
 - **Tier-1:** National centers of supercomputers.
 - **Tier-2:** Regional centers.
- The lower the tier, the larger the applications that can be run.



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

3

3

Cluster Architectures

- Two main types of cluster systems:
 - **Supercomputers:** Organized into modules connected by a network link.
 - Users interact with the front end, which contains software for computations.
 - Jobs are sent to backend modules including compute (CPU), acceleration (GPU), memory (RAM), and storage (HDD).
 - Different modules are prioritized based on the application (deep learning vs. physics simulation).
 - **Load Sharing Facility (LSF):** A network of computers with individual resources (CPU, GPU, RAM, etc.) connected by a network.



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

4

4

Supercomputers vs. LSF Systems

- **LSF systems are better for applications that can be handled by a single node.**
- **Supercomputers are better for resource-intensive applications requiring communication between multiple devices.**



The Role of HPC Schedulers

- **HPC schedulers manage job execution on HPC clusters.**
- They prevent interference between jobs from multiple users on shared resources.
- User applications are queued and run when the requested resources are available.



Popular HPC Schedulers

- SLURM
- MOAB HPC Suite
- PBS Works



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

7

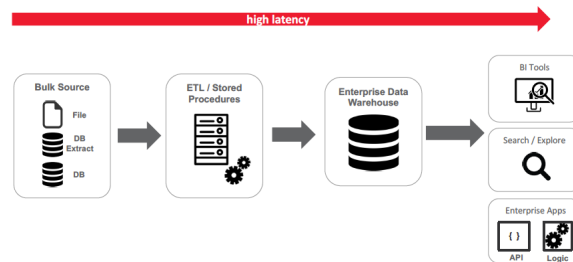
7

Big data architecture

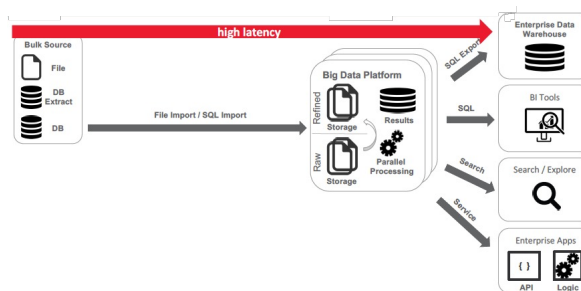
8

8

Traditional BI infrastructures



Hadoop solves Volume and Variety – not Velocity



Lambda Architecture

- A data-processing architecture designed to handle massive quantities of data by taking advantage of both batch and stream processing methods.
- Spark is one of the few data processing frameworks that allows you to seamlessly integrate batch and stream processing
 - Of petabytes of data
 - In the same application

I need fast access
to **historical data**
on the fly for
predictive modeling
with **real time data**
from the **stream**

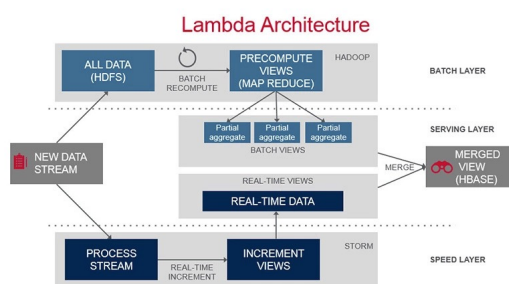


ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

11

11

Lambda architecture



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

12

12

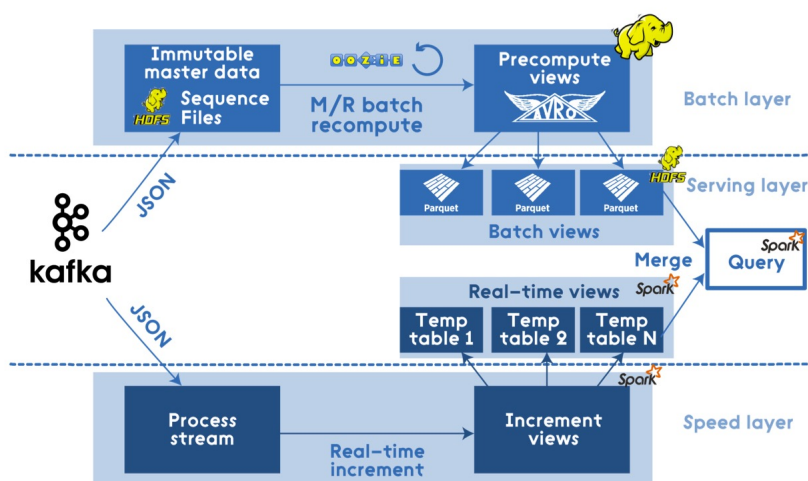
Relevance of data

$\text{query} = \text{function}(\text{batch view}, \text{real time view})$
 $\text{real time view} = \text{function}(\text{real time view}, \text{new data})$
 $\text{batch view} = \text{function}(\text{all data})$



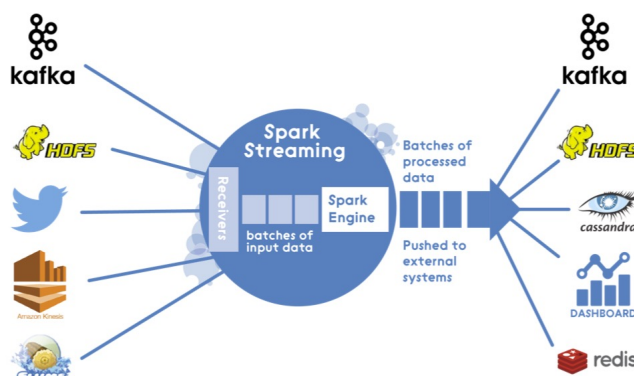
13

Lambda architecture: one implementation



14

Spark streaming



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

15

15

Spark streaming

- Scalable, fault-tolerance stream processing system
- a streaming computation as: a series of very small, deterministic batch jobs
 - Chop up the live stream into batches of X seconds
 - Spark treats each batch of data as RDDs and processes them using RDD operations
 - Finally, the processed results of the RDD operations are returned in batches

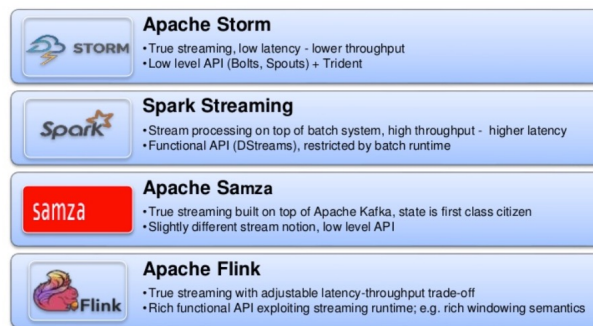


ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

16

16

Streaming landscape

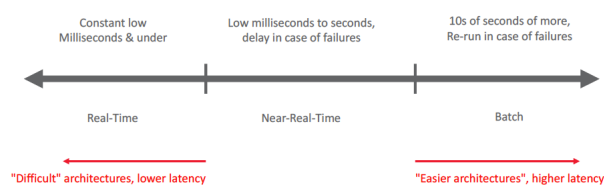


ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

17

17

Stream vs. Batch processing



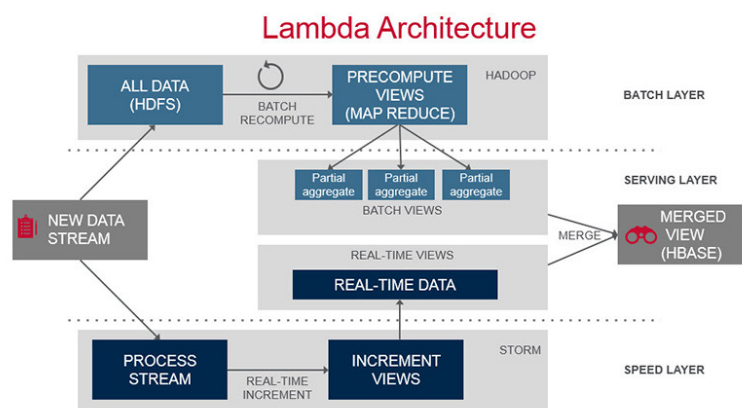
ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

18

18

Case study

Tweets analysis using lambda architecture

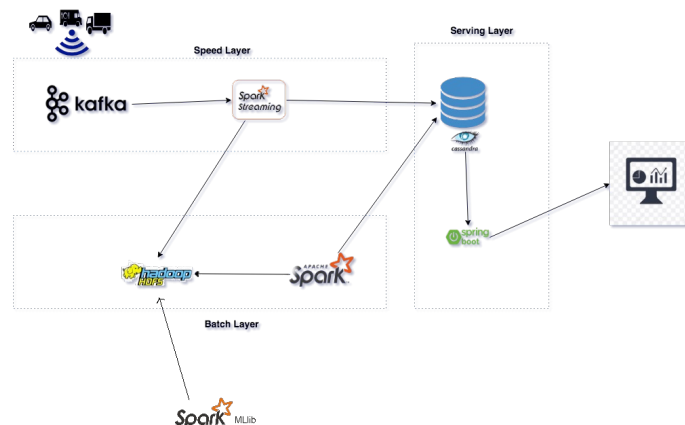


ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

19

Case study

- A full big data pipeline (Lambda Architecture) with Spark, Kafka, HDFS and Cassandra.

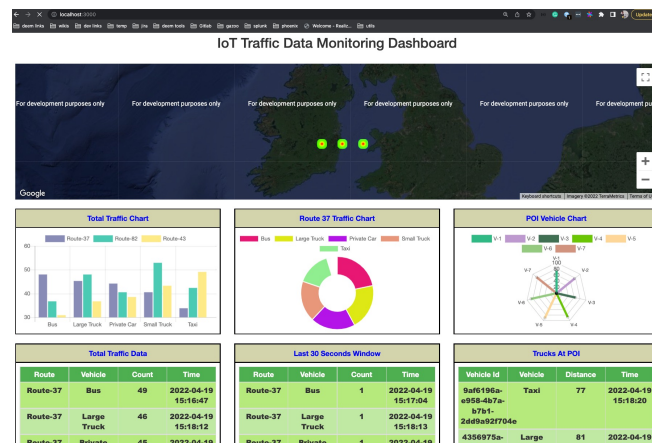


ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

20

Case study

- A full big data pipeline (Lambda Architecture) with Spark, Kafka, HDFS and Cassandra.



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY